

SMART INDIA HACKATHON 2026

GramIntel



Before you borrow, understand what you're building.

- **Problem Statement ID** – 26091
- **Problem Statement Title** – AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs
- **Ministry / Agency** – Ministry of Social Justice & Empowerment (MoSJE)
- **Theme** – Agriculture, FoodTech & Rural Development
- **PS Category** – Software
- **Team ID** – **S-091**
- **Team Name** – **ADAPT**





GramIntel — AI Advisor for Rural Micro-Entrepreneurs



1. Detailed Explanation (Prototype & Core Modules)

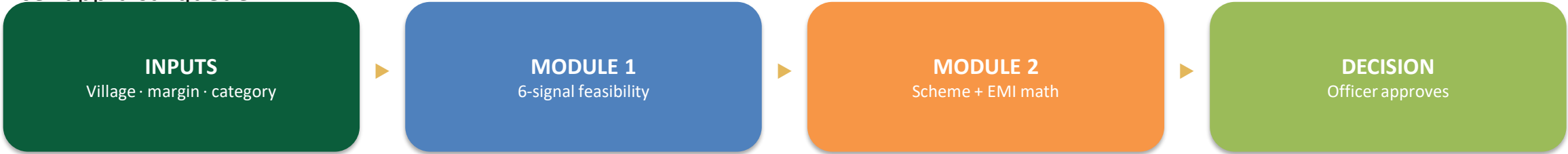
- **Input Parameters:** Village/Block/District location + Available Margin Capital + Proposed Category (Dairy, Retail, Poultry, Food Processing, Textiles, Services, Livestock).
- **Module 1 — Hyper-Local Feasibility:** Evaluates 6 signals (5–10 km market reach, niche opportunity analysis, competitor density, SWOT, threats, and Agmarknet product market value).
- **Module 2 — Smart Scheme Calculator:** 10% margin → 90% loan, auto-routes Micro Finance ($\leq ₹1.40L$ @ 6.5%) vs. Term Loan ($\leq ₹50L$ @ 8%), quarterly EMI/moratorium schedule & 3-mo opex reserve.

2. Addressing the Problem (MoSJE Mandate)

- **Data Over Anecdote:** Replaces hearsay with authentic Census 2011 demographics, Mission Antyodaya infrastructure, and Agmarknet mandi feeds before borrowing.

3. Innovation & Uniqueness

- **Dual-Tier Pipeline & Full Loop:** Multi-tiered fallback (OSM + Census/Antyodaya + Calibrated Baseline) with end-to-end applicant submission to SCA officer appraisal queue.



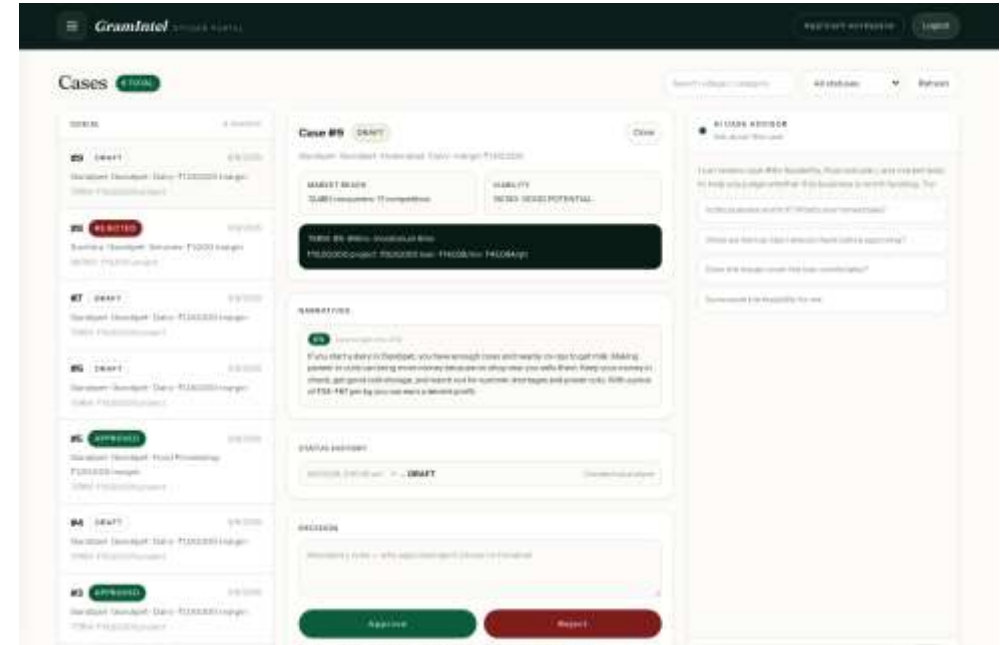
TECHNICAL APPROACH

Technologies Used (4-Layer Modular Stack)

- **Client Layer:** Next.js 14 App Router, React 18, TypeScript 5.7, Tailwind CSS, MapLibre GL 5.6 (pinned), Framer Motion, Lenis.
- **Gateway & Services:** Next.js Proxy, FastAPI, SQLAlchemy/SQLite (zero-ops sync), JWT OTP + Google OAuth with server-side allowlist RBAC.
- **AI & Spatial Pipeline:** Groq LPU (openai/gpt-oss-20b for trilingual narratives & SSE advisor), Overpass API (OSM live shops) with dual-endpoint retries.
- **Data Foundation:** Integrated Census 2011 PCA, Mission Antyodaya 2020 rural infrastructure, and Agmarknet commodity pricing datasets.

Methodology & Implementation Process

- **1. Deterministic Financial Engine:** Deterministic financial & feasibility math strictly verified with 56 automated pytest unit/integration tests.
- **2. Ground-Truth Ingestion:** Directly computes consumer volume and monthly demand from village household counts and category spend benchmarks.
- **3. Multi-Tier Resilience:** Resilient operation with pre-seeded regional fallbacks



56 tests green · next build green · Verified Gov Datasets + PS math

GOV DATASETS & OSM
Census · Antyodaya · Agmarknet

FASTAPI ENGINE
Groq EN/हिं/उ + EMI math

APPS
/assistant + /portal



FEASIBILITY AND VIABILITY



Analysis of Feasibility (Technical, Operational & Financial)

- **Technical Feasibility:** Lightweight commodity stack (Next.js 14 + FastAPI + SQLite), single-image deployment (Vercel + Render), zero-ops maintenance.
- **Operational Feasibility:** Trilingual vernacular interface (EN/हिं/తె) with simple OTP auth, removing digital and language literacy hurdles for rural beneficiaries.
- **Financial Viability:** Aligned with MoSJE/NBCFDC/NSFDC scheme rules; computes 3-mo opex reserve + 10% contingency to safeguard borrower liquidity.

Potential Challenges, Risks & Mitigation Strategies

Risk / Challenge	Mitigation strategy	Status
Unreliable rural connectivity / venue WiFi	Dual-endpoint Overpass retries, localStorage caching, pre-seeded demographic fallbacks; zero blank UI	Built ✓
Data sparsity in remote unsurveyed villages	Grounded in Census 2011 PCA & Mission Antyodaya village infrastructure (co-ops, banks, haats, power)	Built ✓
Groq / External AI service outage	Template narrative fallbacks (EN/हिं/తె) + offline case chat advisor; portal never blocks	Built ✓
Working capital depletion & gestation lag	Built-in financial safety buffer: computes 3-month operating expenses + 10% contingency into project cost	Built ✓

IMPACT AND BENEFITS

Potential Impact on Target Audience

- **Transparent Financial Structuring:** ₹1,00,000 margin → ₹10,00,000 project → ₹9,00,000 loan (TERM @ 8%, ₹14,028/mo EMI) clearly structured before borrowing.
- **Reduces Enterprise Failure Rate:** Entrepreneurs choose viable ventures backed by Census population & Agmarknet demand (e.g. 14,230 pop & ₹53.5L/mo demand in Chevella).

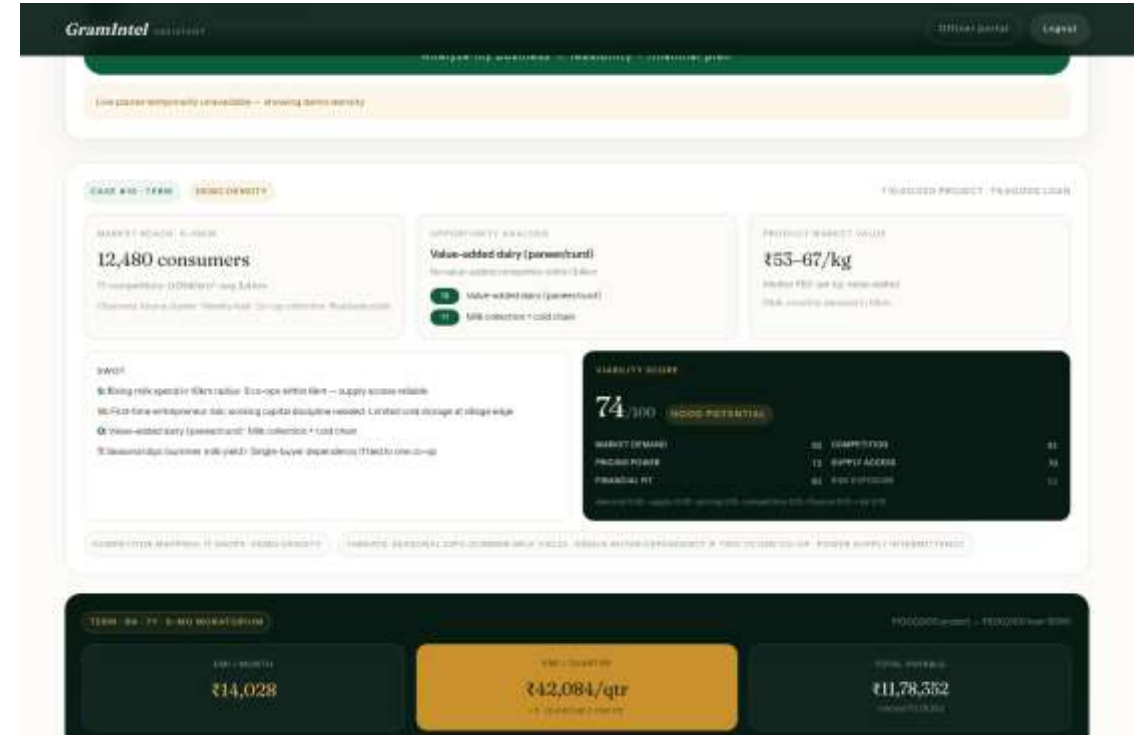
Social, Economic & Governance Benefits

- **Economic (SCA Default Reduction):** Drastically lowers non-performing assets (NPAs) for State Channelising Agencies by directing credit to underserved niches.
- **Social Inclusion & Empowerment:** Empowers rural youth, women artisans (Mahila Samridhi & Swarnima schemes), and grassroots SHGs through vernacular guidance.
- **Governance Acceleration:** Shrinks appraisal turnaround from weeks of paperwork to 4 minutes with streaming AI case advisor & audit history.

Automated Scheme Router (PS-Verbatim Rules):

MICRO ≤ ₹1.40L
6.5% · 3 yr · 3-mo moratorium

TERM ₹1.40–50L
8% · 7 yr · 6-mo moratorium



Live analysis: Census verified · viability 78/100 · quarterly EMI & working capital

RESEARCH AND REFERENCES



- **Scheme Directives (PS-verbatim):** MoSJE, NBCFDC & NSFDC Concessional Loan Guidelines — Micro Finance ($\leq$ ₹1.40L @ 6.5%) & Term Loan ($\leq$ ₹50L @ 8%) — nsfdc.nic.in/scheme · nbcfdc.gov.in.
- **Rural Infrastructure & Demographics:** Ministry of Rural Development (MoRD) — Mission Antyodaya Survey (2020) & Census of India 2011 Primary Census Abstract (PCA) — missionantyodaya.nic.in · censusindia.gov.in.
- **Mandi Pricing & Livestock Benchmarks:** Directorate of Marketing & Inspection (DMI) — Agmarknet Daily Mandi Price Feeds & 20th Livestock Census, DAHD — agmarknet.gov.in · dahd.nic.in.
- **Hyper-Local Spatial Data:** OpenStreetMap Foundation & Overpass API for real-time shop density mapping & dual failover — overpass-api.de · openstreetmap.org.
- **AI Language & Reasoning:** Groq LPU Inference Engine (openai/gpt-oss-20b / llama-3.1-8b) for structured trilingual narratives & SSE streaming advisor — docs.groq.com.
- **Engineering & Verified TDD:** Next.js 14 App Router, FastAPI, SQLAlchemy/SQLite; 56-test pytest suite with golden validation for PS lending boundaries & dataset grounding — github.com/GramIntel.